What Are Multimodal Time-Series Models Actually Reading? Disentangling Text Semantics from Numeric Priors on Time-MMD

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Abstract

Multimodal time-series forecasters now claim that natural-language context substantially improves numeric prediction. Three concurrent paradigms report double-digit MSE gains over unimodal baselines on the Time-MMD benchmark: a pre-trained foundation model (AURORA), an early-fusion text-channel approach (TATS), and a late-fusion residual approach (MM-TSFLIB). We test whether these gains stem from *text semantics* or from a co-shipped numeric *prior column* that flows through the same fusion path. Holding model checkpoints fixed, we perturb Time-MMD along two orthogonal axes – text content (empty, shuffled, cross-domain, constant, oracle) and numeric prior (intact, zeroed) – yielding eight conditions. Across nine domains, four horizons, three seeds, and three backbones per fusion architecture (7,776 **forecasting runs**), we find: (i) zeroing the prior column degrades MSE by up to 31.1%, while every text-only perturbation moves MSE by less than 0.06%; (ii) AURORA's predictions are bit-identical with full text, zeroed text, and the model's own unimodal flag, even though its text encoder still receives non-zero gradients; (iii) AURORA's text cross-attention has near-uniform entropy ($\widehat{H} = 0.975$) regardless of input. On Time-MMD, the apparent multimodal gain is structural rather than semantic, with implications for how benchmarks report multimodal lift and how foundation-model evaluation should be designed.

1. Introduction

Foundation models that consume both numeric time series and natural language are now a fast-growing branch of structured-data research (Liu et al., 2024a; Zhou et al., 2025; Wu et al., 2025; Liu et al., 2024c). On Time-MMD (Liu et al., 2024a) – the standard text–time-series forecasting benchmark – three architectural paradigms have emerged. A zero-shot pretrained model, AURORA (Wu et al., 2025), attends over a tokenised text column. A per-domain trained early-fusion model, TATS (Zhou et al., 2025), projects per-row text embeddings and concatenates them as additional input channels. A per-domain trained late-fusion model, MM-TSFLIB (Liu et al., 2024a), blends a pooled-text vector with the time-series head's output. All three publish double-digit MSE improvements over unimodal baselines.

A closer look at the data pipeline complicates this story. Every Time-MMD domain ships not only paired text but also a numeric column `prior_history_avg`: an LLM-generated forecast for the same horizon (Liu et al., 2024a). Two of the three reference implementations actively use it. In MM-TSFLIB's released training loop the multimodal output is computed as $\texttt{prompt_y} = \texttt{norm}(\texttt{prompt_emb}) + \texttt{prior_y}$, where `prompt_emb` is a frozen-BERT pooled text embedding and `prior_y` is the prior column. TATS adds an analogous output-side blend $\hat{\mathbf{y}} = (1-w)\,\hat{\mathbf{y}}_{\text{model}} + w \cdot \mathbf{p}_{L+1:L+H}$ on top of an early-fusion text channel. AURORA does not consume the prior column, but does cross-attend over the text column. Crucially, in every published implementation, "no text" is operationalised in a way that simultaneously zeroes a numeric signal that has nothing to do with text semantics – a confound that has previously caused well-known reasoning benchmarks to collapse under adversarial probes (Niven & Kao, 2019; McCoy et al., 2019).

We therefore ask: *when these models appear to "use" text, is the gain coming from natural-language semantics, or from the prior column that flows through the same fusion path?* We construct an 8-condition behavioural ladder over two orthogonal axes – text content (intact, empty, within-domain shuffled, cross-domain swapped, constant, oracle pseudo-future) and numeric prior (intact, zeroed) – plus an architectural unimodal switch via each model's published flag. Holding the rest of the pipeline fixed by upstream code with audited patches (Appendix A), we re-run all three architectures across 9 domains, 4 horizons, 3 seeds, and 3 backbones per trained method. For AURORA we addition-

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ally probe the live forward–backward graph: gradient norm at the distilled text tokens, cross-attention entropy at the text guider, and prediction divergence under text ablation (Section 5). Our findings, summarised in Table 2, are blunt.

Contributions. **(1) Reproducibility audit.** We release the first end-to-end unification of AURORA, TATS, and MM-TSFLIB on Time-MMD, with seed-stable data perturbations and bit-level reproducible runs across 7,776 cells. **(2) Two-axis falsification protocol.** Our 8-condition design, formalised as transformations on $(\mathbf{T}, \mathbf{p})$ in Section 4, separates content-bearing text effects from numeric-prior effects within a single architecture and checkpoint. **(3) Mechanistic probes.** For AURORA we show that the text pathway is wired into the gradient graph but its forward-pass contribution is bit-identical to no-text on every cell, with near-uniform cross-attention. **(4) Implications for benchmark design.** We argue that any future multimodal time-series benchmark must report a prior-zeroed control to be informative about text use.

2. Setup and Notation

A Time-MMD dataset is a sequence of date-indexed records. For a univariate forecasting task we organise each record into a sliding window of history length L and forecast horizon H, with N the number of windows in a split. Within a window the four available signals are

$$\mathbf{x} \in \mathbb{R}^L, \quad \mathbf{T} \in \mathcal{T}^L, \quad \mathbf{p} \in \mathbb{R}^{L+H}, \quad \mathbf{y} \in \mathbb{R}^H, \tag{1}$$

where $\mathcal{T}$ is the alphabet of natural-language strings and $\mathbf{T}_t$ is the text written for date t (when no document is available the field carries the empty string ε). The prior column $\mathbf{p}$ is a precomputed LLM-generated forecast included in Time-MMD (Liu et al., 2024a); it is defined for the whole window including the future portion $\mathbf{p}_{L+1:L+H}$, which the fusion paths consume. A multimodal forecaster is then a function $\hat{\mathbf{y}} = f_\theta(\mathbf{x}, \mathbf{T}, \mathbf{p}) \in \mathbb{R}^H$.

A frozen pretrained language model $\phi : \mathcal{T} \to \mathbb{R}^{d_e}$ maps each text row to a d_e-dimensional vector $\phi(\mathbf{T}_t)$. We collect the per-row embeddings into the *text covariate matrix*

$$\mathbf{E}(\mathbf{T}) \;\coloneqq\; \begin{bmatrix} \phi(\mathbf{T}_1) \\ \vdots \\ \phi(\mathbf{T}_L) \end{bmatrix} \in \mathbb{R}^{L \times d_e}. \tag{2}$$

$\mathbf{E}$ is the central object: its row dimension matches the numeric history (so it can be concatenated to $\mathbf{x}$) and its column dimension is the language-model embedding size (so it can be pooled, attended to, or summarised). How an architecture extracts predictive signal from $\mathbf{E}$ is what distinguishes the three paradigms below.

3. Three Multimodal Architectures

TATS (early fusion). ϕ is a frozen GPT-2 pooler (Radford et al., 2019). A trainable MLP $\psi : \mathbb{R}^{d_e} \to \mathbb{R}^{d_p}$ projects each row of $\mathbf{E}$ to a small dimension d_p, after which the projected matrix $\psi(\mathbf{E}) \in \mathbb{R}^{L \times d_p}$ is concatenated to $\mathbf{x}$ along the channel axis and fed to a backbone f_θ (DLinear (Zeng et al., 2023), Informer (Zhou et al., 2021), or iTransformer (Liu et al., 2024b)):

$$\hat{\mathbf{y}}^{\text{TATS}} \;=\; (1-w)\, f_\theta\big(\,[\,\mathbf{x};\ \psi(\mathbf{E})\,]\,\big) \;+\; w\, \mathbf{p}_{L+1:L+H}. \tag{3}$$

The trained text path is the early-fusion concatenation; the prior column enters as an output-side residual blend with weight w tuned per dataset. In the released code, the line that performs the concatenation also calls `.detach()`, severing the gradient into ψ – a structural artefact whose consequences we trace in Appendix F.

MM-TSFLIB (late fusion). ϕ is a frozen BERT pooler (Devlin et al., 2019). The text covariate matrix is summarised to a single horizon-shaped vector by a pooling and small projection, $\bar{\phi}(\mathbf{T}) \in \mathbb{R}^H$, which is then added to the prior horizon and blended residually with the time-series backbone:

$$\hat{\mathbf{y}}^{\text{MM-TSFLIB}} \;=\; (1\text{–}w)\, f_\theta(\mathbf{x}) + w\,\big(\mathrm{norm}(\bar{\phi}(\mathbf{T})) + \mathbf{p}_{L+1:L+H}\big). \tag{4}$$

The pooled-text term and the prior term enter through the *same* additive residual: when $\mathbf{p}$ is zeroed, only the nearly-constant pooled-text contribution remains. This is the mechanistic basis for our C9 (ZERO-PRIORS) probe.

AURORA (cross-attention guidance, zero-shot). ϕ is a frozen BERT producing per-token features $\mathbf{E} \in \mathbb{R}^{L' \times d_e}$ (with $L' = 125$ after sentence-level tokenisation). Two trainable text paths follow: (i) a distillation decoder with $L_k = 10$ learnable query tokens $\mathbf{Q} \in \mathbb{R}^{L_k \times d}$ produces $\mathbf{Z}_T = \mathrm{CrossDec}(\mathbf{Q}, \mathbf{E})$, which biases the encoder's self-attention through a soft gate $\alpha_t = \mathrm{softmax}\big(\mathrm{TextGuider}(\mathbf{x}, \mathbf{Z}_T)\big)$; and (ii) a residual injection of $\mathbf{E}$ into the encoder output through a learned connector `ConnText`. Schematically,

$$\hat{\mathbf{y}}^{\text{AURORA}} \;=\; f_\theta^{\text{AURORA}}\big(\mathbf{x},\ \alpha_t,\ \mathbf{E}\big). \tag{5}$$

AURORA does not consume $\mathbf{p}$ at all; we verified this by inspecting the released model code. By construction AURORA is invariant to C9 – a prediction we use as an internal sanity check, and one that holds bit-exactly (Table 2).

4. The Two-Axis Perturbation Protocol

Existing multimodal time-series papers compare "with text" to "without text", which in light of Equations (3) and (4) conflates two distinct quantities: the text channel's contribution to the forecast, and the prior column's contribution that

Table 1. **Perturbation conditions** as transformations $(\pi_{\text{text}}, \pi_{\text{prior}})$ on $(\mathbf{T}, \mathbf{p})$. σ is a within-domain permutation seeded by the run; $\mathbf{T}'$ is paired cross-domain text aligned to the same date; c is the fixed string "*Time series data point.*"; $\tau(\mathbf{y})$ is the templated oracle string "*The next H values are $y_1, \ldots, y_H$.*" C6 triggers each model's published unimodal flag, which simultaneously drops the text path and the prior fusion.

Cond.	$\pi_{\text{text}}(\mathbf{T})$	$\pi_{\text{prior}}(\mathbf{p})$
C1	$\mathbf{T}$	$\mathbf{p}$
C3	$\mathbf{T}_{\sigma(\cdot)}$	$\mathbf{p}$
C4	$\mathbf{T}'$	$\mathbf{p}$
C8	$\tau(\mathbf{y})$	$\mathbf{p}$
C9	$\mathbf{T}$	$\mathbf{0}$
C2	$(\varepsilon, \ldots, \varepsilon)$	$\mathbf{0}$
C5	$(c, \ldots, c)$	$\mathbf{0}$
C6	architectural unimodal flag	

occurs alongside. We therefore vary $\mathbf{T}$ and $\mathbf{p}$ as independent transformations. Let $\pi_{\text{text}} : \mathcal{T}^L \to \mathcal{T}^L$ act on the text column and $\pi_{\text{prior}} : \mathbb{R}^{L+H} \to \mathbb{R}^{L+H}$ act on the prior column; the data passed to f_θ is then $(\mathbf{x}, \pi_{\text{text}}(\mathbf{T}), \pi_{\text{prior}}(\mathbf{p}))$. Our eight conditions are precisely the values of $(\pi_{\text{text}}, \pi_{\text{prior}})$ shown in Table 1.

Reading the perturbations. Within-domain shuffling (C3) breaks row alignment but preserves the marginal distribution of text at the date level; cross-domain swap (C4) preserves natural-language style but breaks topical relevance; the constant string (C5) removes both content and within-batch variation; and the oracle (C8) injects ground-truth future values into the text column as a templated string, which any text-reading model is forced to benefit from (Niven & Kao, 2019). Following the Time-MMD convention that "no text-derived information" includes both columns, the perturbations C2 and C5 zero $\mathbf{p}$ in addition to perturbing $\mathbf{T}$. Comparing C2/C5/C9 therefore isolates the text channel's contribution *when the prior is held at zero*: same prior treatment, three different text values, predictions expected to differ if and only if text is read.

Why C6 is special. The architectural unimodal switch C6 differs from the data-axis perturbations: it triggers each model's own published unimodal flag. In MM-TSFLIB this sets `prompt_weight` $= 0$, dropping both the pooled-text term and the prior residual in Equation (4) simultaneously. In TATS the flag drops the text concatenation channels and the prior blend. By construction, any C1 $\to$ C6 gap conflates a text contribution and a prior contribution – exactly the conflation our two-axis design separates.

5. Mechanistic Probes for Aurora

AURORA exposes a uniquely clean architecture for mechanistic inspection. We attach three probes to the live forward–backward graph; Appendix C verifies that the hooks are invariance-preserving (forward outputs bit-identical with and without instrumentation).

Probe A – gradient norm at distilled text tokens. $\mathbf{Z}_T \in \mathbb{R}^{B \times L_k \times d}$ from Equation (5) is the trainable hand-off between the frozen-BERT text path and the time-series stream. We measure

$$\widehat{g} \;=\; \sqrt{\mathbb{E}_{(b,k,j)} \left| \left(\nabla_{\mathbf{Z}_T} \mathcal{L}\right)_{b,k,j} \right|^2}, \tag{6}$$

the root-mean-square loss-gradient at the operating point. Hooking $\mathbf{Z}_T$ rather than BERT's input embeddings is required because BERT is frozen – the in-graph trainable text parameters live above this tensor.

Probe B – cross-attention entropy at the text guider. For the cross-attention weights $\alpha_t \in \mathbb{R}^{B \times H \times L_q \times L_k}$ from Equation (5), we compute the per-row Shannon entropy and normalise by the uniform upper bound:

$$\widehat{H} \;=\; \mathbb{E}_{b,h,q}\Big[-\sum_k (\alpha_t)_{bhqk} \log (\alpha_t)_{bhqk} \Big] \,/\, \log L_k. \tag{7}$$

$\widehat{H} = 1$ means uniform attention across the 10 distilled text tokens; values close to 1 indicate the cross-attention does not discriminate between distilled tokens conditional on input.

Probe C – prediction divergence under text ablation. For each batch we forecast with full text and with `text_input_ids=None`, average 10 samples from the flow-matching head per setting, and record the mean squared difference of the two forecasts:

$$\widehat{D} \;=\; \mathbb{E}_b \left\| \hat{\mathbf{y}}_{\text{text}}(b) - \hat{\mathbf{y}}_{\text{none}}(b) \right\|_2^2. \tag{8}$$

$\widehat{D}$ is in standardised target units; a value two orders of magnitude below the test MSE means the text path contributes nothing to the forward pass for this cell.

6. Statistical Protocol

We pair every condition cell with its C1 counterpart at the same (domain, horizon, seed) tuple, compute $\Delta_i = \text{MSE}_i(c) - \text{MSE}_i(\text{C1})$, and obtain 95% confidence intervals via the paired bootstrap (Efron & Tibshirani, 1993) with 10,000 resamples. p-values are two-sided over the bootstrap distribution. All intervals reported in the main tables are computed per-architecture *and per-backbone* on 108 paired cells; full per-backbone breakdowns appear in Appendix H.1.

7. Results

The main results in Table 2 separate the eight conditions into two clean clusters along the prior axis, with the text axis having no detectable effect. Across every backbone, every horizon, and almost every domain, the text-only conditions (C3, C4, C8) sit on top of the original baseline C1 within $\pm 0.06\%$ of MSE, while the prior-zeroed conditions (C9, C2, C5) shift MSE by up to $+31.1\%$ for TATS and $+2.5\%$ for MM-TSFLIB. Even the oracle condition C8, which embeds the ground-truth future as a templated text string, fails to improve any of the three models: the largest move is AURORA's $+0.024\%$ ($p = 0.337$). Bootstrap confidence intervals are tight and either straddle zero or move only in the third decimal of MSE.

C9, C2, and C5 produce *numerically identical* MSEs on TATS and MM-TSFLIB: same prior treatment, three different text values (real, empty, constant), no detectable difference in forecasts. AURORA is unaffected by C9 because $\mathbf{p}$ does not appear in Equation (5), and is moved by less than 0.03% under C2/C5. The two-axis design therefore *localises* the published "multimodal lift" inside a numeric residual that has no semantic-text content.

AURORA is content-blind. Across the 108 cells in our AURORA grid, MSE for C1, C9, and C6 is bit-identical to 13 significant figures. Table 3 reports the three probes (Equations (6) to (8)). Probe A confirms the text pathway is wired in – gradients reach the distilled text tokens with magnitudes $\widehat{g} \in [0.07,\ 0.27]$ across conditions – but Probe B reports $\widehat{H} = 0.975 \pm 0.01$ regardless of input (the cross-attention is essentially uniform over the 10 distilled text tokens), and Probe C reports $\widehat{D} \leq 0.06$ everywhere, two orders of magnitude below test-MSE scale. The text pathway is computed but its forward-pass contribution is null.

8. Discussion

The published unimodal gap is the prior gap. In the original TATS and MM-TSFLIB papers, the headline result is the C1 − C6 gap: $+5.24\%$ for TATS and $+1.76\%$ for MM-TSFLIB in our re-runs. But C6 drops both the text channel *and* the numeric-prior fusion (Section 4). Comparing with C9, which drops only the numeric prior, we see $+31.06\%$ and $+2.44\%$ respectively – already accounting for and exceeding the headline gap. The published gap is attributable to numeric-prior fusion, not to text semantics.

Why does AURORA still claim multimodal gains? AURORA never reads $\mathbf{p}$, yet its public results report a small C1–C6 improvement on Time-MMD. Our re-runs reproduce that small effect. But the text pathway in the released checkpoint has near-uniform cross-attention and zero forward-pass effect on the test grid, even though gradients flow. A trained text path that is not content-discriminative is structurally similar to a frozen no-op: it cannot worsen predictions, and on Time-MMD it cannot improve them.

Architectural archaeology. TATS's training loop concatenates the projected text embedding with the numeric input under a `.detach()` call (line 519 of `exp_long_term_forecasting.py`), severing the gradient into ψ even though an optimiser step on ψ is invoked every iteration. Combined with the residual blend in Equation (3), this means a TATS run on Time-MMD is structurally incapable of learning to use text for prediction beyond random projection variance. Full code-level audit in Appendix F.

Limitations. We test three architectures. AURORA's pretraining corpus is large and only partially documented; we cannot rule out Time-MMD contamination at pretraining time (Appendix G). Our perturbation set covers row-level text variations; we do not vary the LLM that generated $\mathbf{p}$. All claims are about Time-MMD; the generality of the behavioural ladder to other text–time-series benchmarks is an open question.

Implications for benchmark design. A multimodal time-series benchmark whose "text" channel is read by the model only through a residual that also consumes a numeric column will report "multimodal" lifts even if the text is empty. This echoes parallel critical evaluations of (unimodal) time-series foundation models (Zhang et al., 2025): small, careful baselines and falsification controls often outperform expensive multimodal machinery on the very benchmarks built for them. We release our perturbation generator, runner harnesses, and probe instrumentation as a drop-in evaluation suite, and we recommend that any future text–time-series benchmark report a prior-zeroed control (C9) and an empty-text control (C2) alongside any "multimodal" headline.

9. Conclusion

The three publicly released multimodal time-series architectures on Time-MMD appear to use natural-language context. Under a two-axis perturbation protocol with 7,776 controlled runs and three mechanistic probes for the foundation-model entry, we find that text content moves MSE by less than 0.06% in every condition; that the single perturbation that does move MSE is zeroing a numeric prior column unrelated to language; and that AURORA's text pathway has near-uniform cross-attention and produces bit-identical predictions with and without text on the entire test grid. We release the benchmark suite to make these controls a default rather than an afterthought.

Table 2. **Main results.** Mean test MSE on Time-MMD per (model, condition, domain), averaged across 4 horizons, 3 seeds, and (for trained methods) 3 canonical backbones (DLinear, Informer, iTransformer). **Bold** marks the lowest MSE per (model×column); underline marks the second lowest. The column **Avg** is the cross-domain mean. C9, C2, and C5 are the *prior-zeroed* conditions; C3, C4, and C8 are the *text-only* perturbations; see Table 1 for the formal definitions.

Model	Condition	Agri	Clim	Econ	Ener	Envi	Heal	Secu	SocG	Traf	Avg
AURORA	C1 Original	**0.275**	**0.865**	0.033	**0.255**	0.276	1.553	72.72	0.836	0.161	8.553
	C3 Shuffled	0.275	0.865	0.034	0.255	0.276	1.553	72.72	**0.835**	0.161	8.553
	C4 Cross-domain	0.275	0.865	0.033	0.255	0.276	1.552	**72.72**	0.838	**0.161**	**8.552**
	C8 Oracle	0.275	0.865	0.035	0.255	0.276	1.557	72.73	0.844	0.163	8.555
	C9 Zero-priors	**0.275**	**0.865**	0.033	**0.255**	0.276	1.553	72.72	0.836	0.161	8.553
	C2 Empty	0.275	0.865	**0.032**	0.256	**0.276**	**1.551**	72.74	0.836	0.161	8.555
	C5 Constant	0.277	0.865	0.035	0.255	0.276	1.559	72.73	0.845	0.163	8.555
	C6 Unimodal	**0.275**	**0.865**	0.033	**0.255**	0.276	1.553	72.72	0.836	0.161	8.553
MM-TSFLIB	C1 Original	0.231	1.163	**0.359**	0.345	0.477	1.575	**118.7**	1.040	0.234	13.79
	C3 Shuffled	0.227	**1.158**	0.372	0.345	0.477	1.575	118.8	1.037	0.234	13.80
	C4 Cross-domain	**0.223**	1.161	0.371	**0.343**	0.477	**1.565**	118.7	**1.033**	**0.234**	**13.79**
	C8 Oracle	0.234	1.163	0.386	0.345	0.479	1.574	118.8	1.056	0.235	13.80
	C9 Zero-priors	0.800	1.258	0.420	0.407	0.473	1.707	120.6	1.166	0.369	14.13
	C2 Empty	0.799	1.262	0.441	0.402	**0.473**	1.711	120.6	1.165	0.371	14.14
	C5 Constant	0.804	1.261	0.437	0.402	0.474	1.718	120.6	1.164	0.373	14.14
	C6 Unimodal	0.270	1.249	0.457	0.349	0.502	1.643	120.5	1.078	0.270	14.04
TATS	C1 Original	**0.162**	0.967	0.127	0.387	0.290	1.333	**115.3**	1.029	0.189	**13.31**
	C3 Shuffled	0.163	0.967	**0.127**	0.387	0.290	1.333	115.3	1.029	**0.189**	13.31
	C4 Cross-domain	0.162	**0.967**	0.127	0.388	**0.290**	1.333	115.3	1.029	0.189	13.31
	C8 Oracle	0.163	0.967	0.127	0.388	0.290	**1.332**	115.3	**1.029**	0.189	13.31
	C9 Zero-priors	15.99	3.317	0.426	1.435	0.407	2.009	125.9	3.290	4.212	17.44
	C2 Empty	15.99	3.317	0.426	1.435	0.406	2.009	125.9	3.290	4.212	17.44
	C5 Constant	15.99	3.317	0.426	1.435	0.407	2.009	125.9	3.290	4.212	17.44
	C6 Unimodal	0.281	1.250	0.517	**0.345**	0.428	1.691	120.2	1.077	0.289	14.00

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Table 3. **AURORA probes** (Equations (6) to (8)) across the eight conditions. $\widehat{g}$ is the gradient norm at the distilled text tokens (Probe A); $\widehat{H}$ is the relative cross-attention entropy (Probe B; $1 =$uniform over $L_k = 10$ tokens); $\widehat{D}$ is the prediction divergence under text ablation (Probe C). Mean $\pm$ s.d. over $n = 27$ probe cells. The cross-attention entropy is 0.975–0.976 in every condition, indistinguishable from uniform.

Cond.	$\widehat{g}$	$\widehat{H}$	$\widehat{D}$
C1	0.152 ± 0.24	$0.975 \pm .009$	$0.041 \pm .037$
C3	0.155 ± 0.40	$0.975 \pm .009$	$0.035 \pm .035$
C4	0.072 ± 0.08	$0.975 \pm .009$	$0.035 \pm .032$
C8	0.133 ± 0.19	$0.976 \pm .009$	$0.058 \pm .084$
C9	0.265 ± 0.67	$0.975 \pm .009$	$0.028 \pm .018$
C2	0.074 ± 0.10	$0.975 \pm .009$	$0.040 \pm .041$
C5	0.156 ± 0.23	$0.976 \pm .009$	$0.044 \pm .035$
C6	0.080 ± 0.15	$0.975 \pm .009$	$0.035 \pm .024$

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A. Upstream Patches and Reproducibility Audit

We pin every upstream repository to a specific commit and apply 13 small patches via `code/apply_repo_patches.py`. Patches are mechanically diffable; this section summarises the categories and their consequences for reproducibility.

Data-loading correctness. Time-MMD ships rows whose textual columns are sometimes empty strings. pandas's default `read_csv` converts these to `NaN`, which several upstream loaders then silently replace with 0 or with the string `"nan"` – both of which destroy the meaning of our C2 (EMPTY) condition. We pass `keep_default_na=False` in every `read_csv` call in the three repos, and we remove a code path in MM-TSFLIB that `fillna("nan")`s the text column.

API drift. All three repos were released against pandas 1.x. We backport four idioms (`drop(columns=)` instead of the deprecated positional second argument, `Series.iloc[]` casting, etc.) so the same code path runs on pandas 2.x without behavioural changes.

Architectural unimodal switch. TATS and MM-TSFLIB ship CLI flags for unimodal evaluation (`--text_emb 0 --prior_weight 0` and `--prompt_weight 0` respectively). AURORA hard-codes the text path to always run. We add a `--no_text` flag to AURORA's run script that sets `text_input_ids=None` before calling the model's `generate`. By construction this is the only path the model exposes for skipping the text branch; Appendix C verifies it matches default-text behaviour bit-exactly on every Time-MMD cell.

Backbone registration. TATS's `exp/exp_basic.py` registers only ITRANSFORMER. We register the eight additional backbones present in the same source tree (AUTOFORMER, DLINEAR, INFORMER, FILM, FEDFORMER, PATCHTST, CROSSFORMER, TRANSFORMER).

Determinism and seeding. Each runner sets the model's RNG via the published seed flag. We set `torch.manual_seed` and `numpy.random.seed` per cell. The orchestrator also fixes `CUDA_VISIBLE_DEVICES` per shard; we did not pin `torch.use_deterministic_algorithms(True)` because that disagrees with one of MM-TSFLIB's flash-attention paths. AURORA's flow-matching head averages 100 samples for inference; with seeded sampling we obtain bit-identical predictions per cell across re-runs (Appendix E).

Reproducibility. The end-to-end sweep is sharded across GPUs and resumable; total wall clock for the 7,776-cell sweep is $\sim$ 92 A10G-hours. The full log of CLI invocations, git SHAs, and returned MSE/MAE per cell is in our `sweep_log.jsonl` (released alongside the code).

B. Perturbation Generation Details

Row-level invariants. For all data-axis perturbations (C2–C5, C8, C9) we preserve the row count, the date column, and the numeric target column. Our generator `code/generate_perturbations.py` runs a post-hoc validator that diffs every perturbed CSV against the original on the columns that should be unchanged.

C3 (SHUFFLED). We permute the text column within the same domain using a NumPy random generator seeded by the RunSpec seed. This is the only condition whose CSV content depends on the seed; all other conditions live under `seed0/` on disk.

C4 (CROSS-DOMAIN). We use a fixed pairing f:AGRICULTURE$\rightarrow$CLIMATE, CLIMATE$\rightarrow$AGRICULTURE, and so on. For each target row we pull the text from the source-domain row whose `date` is closest to the target row's date (resolving ties deterministically). This preserves natural English while breaking topic alignment.

C8 (ORACLE). For each row in the test window we substitute a templated string of the form "`The next` H `values are` $y_1, y_2, \ldots, y_H$`.`" using the ground-truth values. The substitution is applied to train/val rows – rolled back by H steps – so the model sees consistent oracle structure during fine-tuning (for TATS and MM-TSFLIB). For AURORA the test window is what matters.

C9 (ZERO-PRIORS). We set every entry of `prior_history_avg` to 0.0. AURORA does not consume this column at all (verified by `grep -n prior_history Aurora/`), so C9 is by construction a no-op for AURORA – exactly the behaviour observed to bit precision.

Why C2 and C5 also zero the prior. Following the original Time-MMD convention, the perturbations that strip "text-derived information" do so for all text-derived numeric columns as well, including `prior_history_avg`. This is what makes C2/C5/C9 a clean three-way comparison: prior is held at zero in all three, and text varies between empty, constant, and intact. If the model reads text at all, these three should differ.

Examples. Table 4 shows one example row from each condition for the PUBLIC_HEALTH domain (week of 2013-02-04, weighted ILI percentage forecasting).

C. Probe Output Invariance

We run AURORA's `generate` call twice on a fixed batch with the same seed: once with no hooks, once with both gradient-norm and attention-entropy hooks attached. Forward outputs match bit-exactly (`torch.allclose(rtol=0,atol=0)`); the hook attachment does not perturb the forward graph. Code in `code/aurora_probes.py::validate_invariance`.

D. Per-backbone Bootstrap Confidence Intervals

A condensed view is given in Table 5, which reports paired-bootstrap means and 95% confidence intervals against C1, broken out per backbone. All intervals use 10,000 resamples on 108 paired cells per (model, backbone, condition). The cluster structure of Table 2 is preserved across every backbone: text-content perturbations (C3, C4, C8) have intervals tight to zero, while prior-zeroing perturbations (C2, C5, C9) have intervals visibly above zero.

A full per-(model, backbone, condition) breakdown – including all nine conditions and including MM-TSFLIB/iTransformer and the three TATS backbones – is provided in Appendix H.

E. Bit-Identical Cells: Direct Verification

For AURORA, $108/108$ paired cells satisfy $\mathrm{MSE}(\mathrm{C1}) = \mathrm{MSE}(\mathrm{C9}) = \mathrm{MSE}(\mathrm{C6})$ to 13 significant figures; the bootstrap interval is $[0, 0]$ exactly because every paired difference is exactly zero. For TATS on DLINEAR and ITRANSFORMER, C2, C5, and C9 give identical predicted forecasts cell-by-cell because of the architectural artefact described in Appendix F; these match to ~ 13 significant figures of MSE rather than to bit precision (small floating-point non-associativity in the forward pass).

F. Architectural Diagnostic: TATS's `detach()` Edge

In TATS's training loop (`repos/TaTS/exp/exp_long_term_forecasting.py:519`):

```
batch_x = torch.cat(
    [batch_x, prompt_emb], dim=-1
).detach()
```

The `.detach()` severs gradient flow into `prompt_emb`, which is the output of `self.mlp(text_embeddings)`. TATS's training loop nevertheless instantiates a separate optimiser for `self.mlp` and calls `model_optim_mlp.step()` every iteration – but those parameters never receive gradients.

Consequence. The text-projection MLP ψ stays at random initialisation throughout fine-tuning. The downstream backbone receives a random projection of frozen-LLM features as additional input channels; over training it learns to assign small or zero weight to those channels because they cannot reduce loss reliably. At test time, with text content varied (C2/C5/C9) the random projection maps to slightly different vectors but the trained backbone has already collapsed those channels to near-zero contribution. This is a sufficient mechanical explanation for the bit-identical behaviour we observe in Table 2.

Significance. This is not a takedown of TATS as a research artefact; the released code may differ from the authors' internal codebase. We highlight it because anyone re-deriving a "multimodal" lift from TATS's public code is, structurally, re-deriving the prior-residual gap rather than a text gap.

G. What about Aurora's Pretraining?

AURORA's zero-shot regime means our findings concern the released *checkpoint*'s use of text on Time-MMD. We do not re-pretrain AURORA; we cannot rule out training-time contamination of Time-MMD content into the pretraining corpus. The fact that $\mathrm{C1}/\mathrm{C9}/\mathrm{C6}$ are bit-identical and $\mathrm{C2}/\mathrm{C5}$ are within 0.03% of C1 tells us that, conditional on the re-

Table 4. **Concrete examples** of the text and prior values fed to the model under each condition for one fixed row of the PUBLIC_HEALTH domain (US_FLURATIO_WEEK, week of 2013-02-04, ground-truth OT = 3.236). "Text" is the `Final_Search_4` column (the one consumed by the fusion path); "Prior" is `prior_history_avg`. Texts truncated at ∼100 characters. By design, C1 = C9 in text and C2 = C5 = C9 in prior, so each pair within those groups isolates the effect of varying the *other* axis.

Cond.	Text fed to fusion path	Prior
C1	"Available facts are as follows: 2013-01-28: Warm winters tend to be followed by severe influenza epidemics, and climate…"	2.854
C3	"Available facts are as follows: 2005-03-21: Objective facts about the Public Health and FLU situation: There have been…"	2.854
C4	"Available facts are as follows: 2013-01-21: The U.S. trade balance has deteriorated by more than $8 billion from 1964 to 1971…"	2.854
C8	"Available facts are as follows: Step+1: The OT will be 2.894210. Step+2: The OT will be 2.717420. Step+3: The OT will…"	2.854
C9	"Available facts are as follows: 2013-01-28: Warm winters tend to be followed by severe influenza epidemics, and climate…"	0.000
C2	*(empty string)*	0.000
C5	"Time series data point."	0.000

Table 5. Mean ΔMSE vs. C1 with 95% paired bootstrap CI (n = 108 paired cells, 10,000 resamples). Statistically-significant differences ($p < 0.01$) marked ⋆. Source: `summaries/bootstrap_per_backbone.csv`.

Model/Bbone	Cond.	$\Delta\overline{\text{MSE}}$	95% CI
AURORA	C2	0.00202 ⋆	[+0.0003, +0.0041]
	C3	0.00032	[−0.00007, +0.0008]
	C4	−0.00051	[−0.0028, +0.0016]
	C9	0	[0, 0]
MM-TSFLIB/DLin	C2	0.297 ⋆	[+0.227, +0.374]
	C3	0.00071 ⋆	[+0.0002, +0.0013]
	C9	0.298 ⋆	[+0.227, +0.374]
	C6	0.331 ⋆	[+0.201, +0.480]
MM-TSFLIB/Inf	C2	0.204 ⋆	[+0.120, +0.303]
	C3	0.018	[+0.001, +0.037]
	C9	0.211 ⋆	[+0.123, +0.301]
	C6	0.201 ⋆	[+0.121, +0.292]

leased checkpoint, text content is not a behavioural lever on this benchmark. Our claim is conditional on these checkpoints; we do not claim that no text–time-series foundation model can use text.

H. Per-backbone, Per-domain, Per-horizon Tables

The next three subsections expand Table 2 along three axes: (i) per backbone, (ii) per domain, and (iii) per horizon. In every breakdown, the two-cluster pattern from the main paper is preserved. Tables are auto-generated from the locked summaries by `paper/figures/make_appendix_tables.py`.

H.1. Per-backbone

Table 6. Per-backbone mean MSE on Time-MMD for AURORA. Each cell is the mean over 9 domains × 4 horizons × 3 seeds. Δ% vs. C1 reported in parentheses. Source: `summaries/per_backbone_aurora.csv`.

Cond.	default
C1	8.553
C3	8.553 (+0.00%)
C4	8.552 (-0.01%)
C8	8.555 (+0.02%)
C9	8.553 (+0.00%)
C2	8.555 (+0.02%)
C5	8.555 (+0.03%)
C6	8.553 (+0.00%)

H.2. Per-domain

H.3. Per-horizon

The horizons available depend on the source domain's sampling frequency: ENVIRONMENT (daily) uses $\{48, 96, 192, 336\}$, HEALTH/ENERGY (weekly) use $\{12, 24, 36, 48\}$, and the remaining six monthly domains use $\{6, 8, 10, 12\}$. The table below merges across domains, so columns with a particular horizon H aggregate over only the domains that include that horizon.

Table 7. Per-backbone mean MSE on Time-MMD for MM-TSFLIB. Each cell is the mean over 9 domains $\times 4$ horizons $\times 3$ seeds. $\Delta\%$ vs. C1 reported in parentheses. Source: `summaries/per_backbone_mmtsflib.csv`.

Cond.	DLinear	Informer	iTransformer
C1	12.755	15.065	13.565
C3	12.755 (+0.01%)	15.082 (+0.12%)	13.558 (-0.05%)
C4	12.755 (+0.01%)	15.075 (+0.06%)	13.553 (-0.09%)
C8	12.756 (+0.01%)	15.087 (+0.15%)	13.566 (+0.01%)
C9	13.052 (+2.33%)	15.263 (+1.32%)	14.080 (+3.80%)
C2	13.052 (+2.33%)	15.269 (+1.36%)	14.084 (+3.83%)
C5	13.052 (+2.33%)	15.276 (+1.40%)	14.085 (+3.83%)
C6	13.086 (+2.60%)	15.266 (+1.33%)	13.762 (+1.45%)

Table 8. Per-backbone mean MSE on Time-MMD for TATS. Each cell is the mean over 9 domains $\times 4$ horizons $\times 3$ seeds. $\Delta\%$ vs. C1 reported in parentheses. Source: `summaries/per_backbone_tats.csv`.

Cond.	DLinear	Informer	iTransformer
C1	12.421	14.185	13.316
C3	12.421 (+0.00%)	14.186 (+0.01%)	13.316 (-0.00%)
C4	12.421 (-0.00%)	14.186 (+0.01%)	13.316 (-0.00%)
C8	12.421 (-0.00%)	14.185 (+0.00%)	13.316 (-0.00%)
C9	18.578 (+49.57%)	15.257 (+7.56%)	18.486 (+38.82%)
C2	18.578 (+49.57%)	15.257 (+7.56%)	18.486 (+38.82%)
C5	18.578 (+49.57%)	15.256 (+7.55%)	18.486 (+38.82%)
C6	13.165 (+5.99%)	15.381 (+8.43%)	13.467 (+1.13%)

Table 9. Per-domain mean MSE on Time-MMD for AURORA. Each cell is the mean over 4 horizons × 3 seeds (and 3 backbones for trained methods). Δ% vs. C1 in parentheses. Source: `summaries/per_domain_aurora.csv`.

Cond.	Agricul	Climate	Economy	Energy	Environ	Health	Securit	SocialG	Traffic
C1	0.275	0.865	0.033	0.255	0.276	1.553	72.722	0.836	0.161
C3	0.275 (+0.01%)	0.865 (+0.01%)	0.034 (+2.21%)	0.255 (+0.01%)	0.276 (+0.00%)	1.553 (+0.00%)	72.725 (+0.00%)	0.835 (-0.08%)	0.161 (+0.20%)
C4	0.275 (+0.09%)	0.865 (+0.02%)	0.033 (+0.78%)	0.255 (+0.02%)	0.276 (+0.00%)	1.552 (-0.05%)	72.716 (-0.01%)	0.838 (+0.32%)	0.161 (-0.25%)
C8	0.275 (+0.04%)	0.865 (+0.02%)	0.035 (+4.13%)	0.255 (+0.25%)	0.276 (+0.00%)	1.557 (+0.27%)	72.725 (+0.00%)	0.844 (+0.93%)	0.163 (+0.88%)
C9	0.275 (+0.00%)	0.865 (+0.00%)	0.033 (+0.00%)	0.255 (+0.00%)	0.276 (+0.00%)	1.553 (+0.00%)	72.722 (+0.00%)	0.836 (+0.00%)	0.161 (+0.00%)
C2	0.275 (+0.09%)	0.865 (+0.01%)	0.032 (-3.47%)	0.256 (+0.36%)	0.276 (-0.04%)	1.551 (-0.10%)	72.743 (+0.03%)	0.836 (-0.03%)	0.161 (-0.20%)
C5	0.277 (+0.64%)	0.865 (+0.03%)	0.035 (+4.70%)	0.255 (+0.05%)	0.276 (+0.06%)	1.559 (+0.39%)	72.725 (+0.00%)	0.845 (+1.05%)	0.163 (+1.05%)
C6	0.275 (+0.00%)	0.865 (+0.00%)	0.033 (+0.00%)	0.255 (+0.00%)	0.276 (+0.00%)	1.553 (+0.00%)	72.722 (+0.00%)	0.836 (+0.00%)	0.161 (+0.00%)

Table 10. Per-domain mean MSE on Time-MMD for MM-TSFLIB. Each cell is the mean over 4 horizons × 3 seeds (and 3 backbones for trained methods). Δ% vs. C1 in parentheses. Source: `summaries/per_domain_mmtsflib.csv`.

Cond.	Agricul	Climate	Economy	Energy	Environ	Health	Securit	SocialG	Traffic
C1	0.231	1.163	0.359	0.345	0.477	1.575	118.729	1.040	0.234
C3	0.227 (-1.64%)	1.158 (-0.42%)	0.372 (+3.56%)	0.345 (-0.02%)	0.477 (+0.00%)	1.575 (+0.05%)	118.761 (+0.03%)	1.037 (-0.32%)	0.234 (+0.06%)
C4	0.223 (-3.35%)	1.161 (-0.11%)	0.371 (+3.33%)	0.343 (-0.47%)	0.477 (-0.03%)	1.565 (-0.62%)	118.741 (+0.01%)	1.033 (-0.68%)	0.234 (-0.09%)
C8	0.234 (+1.35%)	1.163 (-0.00%)	0.386 (+7.40%)	0.345 (+0.07%)	0.479 (+0.28%)	1.574 (-0.01%)	118.753 (+0.02%)	1.056 (+1.53%)	0.235 (+0.57%)
C9	0.800 (+246.50%)	1.258 (+8.21%)	0.420 (+17.00%)	0.407 (+18.06%)	0.473 (-0.89%)	1.707 (+8.40%)	120.586 (+1.56%)	1.166 (+12.13%)	0.369 (+57.56%)
C2	0.799 (+245.84%)	1.262 (+8.55%)	0.441 (+22.74%)	0.402 (+16.68%)	0.473 (-1.00%)	1.711 (+8.67%)	120.593 (+1.57%)	1.165 (+11.98%)	0.371 (+58.37%)
C5	0.804 (+248.41%)	1.261 (+8.46%)	0.437 (+21.56%)	0.402 (+16.58%)	0.474 (-0.71%)	1.718 (+9.12%)	120.603 (+1.58%)	1.164 (+11.96%)	0.373 (+59.43%)
C6	0.270 (+16.97%)	1.249 (+7.43%)	0.457 (+27.11%)	0.349 (+1.22%)	0.502 (+5.20%)	1.643 (+4.33%)	120.523 (+1.51%)	1.078 (+3.68%)	0.270 (+15.55%)

Table 11. Per-domain mean MSE on Time-MMD for TATS. Each cell is the mean over 4 horizons × 3 seeds (and 3 backbones for trained methods). Δ% vs. C1 in parentheses. Source: `summaries/per_domain_tats.csv`.

Cond.	Agricul	Climate	Economy	Energy	Environ	Health	Securit	SocialG	Traffic
C1	0.162	0.967	0.127	0.387	0.290	1.333	115.282	1.029	0.189
C3	0.163 (+0.11%)	0.967 (-0.02%)	0.127 (-0.04%)	0.387 (+0.01%)	0.290 (-0.10%)	1.333 (-0.01%)	115.285 (+0.00%)	1.029 (+0.02%)	0.189 (-0.02%)
C4	0.162 (+0.07%)	0.967 (-0.02%)	0.127 (-0.03%)	0.388 (+0.03%)	0.290 (-0.13%)	1.333 (-0.01%)	115.285 (+0.00%)	1.029 (-0.01%)	0.189 (-0.02%)
C8	0.163 (+0.11%)	0.967 (-0.00%)	0.127 (-0.01%)	0.388 (+0.03%)	0.290 (-0.00%)	1.332 (-0.03%)	115.283 (+0.00%)	1.029 (-0.01%)	0.189 (+0.16%)
C9	15.990 (+9749.88%)	3.317 (+242.95%)	0.426 (+235.06%)	1.435 (+270.30%)	0.407 (+40.37%)	2.009 (+50.73%)	125.877 (+9.19%)	3.290 (+219.74%)	4.212 (+2127.64%)
C2	15.990 (+9749.88%)	3.317 (+242.95%)	0.426 (+235.02%)	1.435 (+270.31%)	0.406 (+40.15%)	2.009 (+50.76%)	125.877 (+9.19%)	3.290 (+219.73%)	4.212 (+2127.65%)
C5	15.990 (+9749.89%)	3.317 (+242.95%)	0.426 (+235.20%)	1.435 (+270.29%)	0.407 (+40.18%)	2.009 (+50.75%)	125.873 (+9.19%)	3.290 (+219.74%)	4.212 (+2127.64%)
C6	0.281 (+73.02%)	1.250 (+29.22%)	0.517 (+307.00%)	0.345 (-10.90%)	0.428 (+47.44%)	1.691 (+26.85%)	120.161 (+4.23%)	1.077 (+4.63%)	0.289 (+53.02%)

Table 12. Per-horizon mean MSE for AURORA on Time-MMD. Aggregated over domains, seeds, and (for trained methods) backbones; Δ% vs. C1 at the same horizon in parentheses.

Cond.	$H = 6$	$H = 8$	$H = 10$	$H = 12$	$H = 24$	$H = 36$	$H = 48$	$H = 96$	$H = 192$	$H = 336$
C1	11.556	12.144	12.866	10.174	0.897	0.991	0.841	0.284	0.270	0.269
C3	11.557 (+0.01%)	12.145 (+0.01%)	12.866 (+0.00%)	10.173 (-0.00%)	0.897 (+0.00%)	0.991 (-0.01%)	0.841 (+0.00%)	0.284 (+0.00%)	0.270 (+0.00%)	0.269 (+0.00%)
C4	11.563 (+0.06%)	12.147 (+0.02%)	12.860 (-0.04%)	10.169 (-0.05%)	0.897 (-0.01%)	0.990 (-0.06%)	0.841 (-0.03%)	0.284 (+0.00%)	0.270 (-0.00%)	0.269 (-0.00%)
C8	11.573 (+0.14%)	12.152 (+0.07%)	12.862 (-0.03%)	10.165 (-0.09%)	0.899 (+0.22%)	0.994 (+0.33%)	0.843 (+0.28%)	0.284 (+0.00%)	0.270 (+0.00%)	0.269 (+0.01%)
C9	11.556 (+0.00%)	12.144 (+0.00%)	12.866 (+0.00%)	10.174 (+0.00%)	0.897 (+0.00%)	0.991 (+0.00%)	0.841 (+0.00%)	0.284 (+0.00%)	0.270 (+0.00%)	0.269 (+0.00%)
C2	11.554 (-0.01%)	12.145 (+0.01%)	12.871 (+0.05%)	10.179 (+0.05%)	0.897 (+0.03%)	0.990 (-0.08%)	0.840 (-0.04%)	0.284 (-0.04%)	0.270 (-0.05%)	0.269 (-0.06%)
C5	11.576 (+0.17%)	12.154 (+0.08%)	12.861 (-0.03%)	10.164 (-0.10%)	0.899 (+0.27%)	0.996 (+0.46%)	0.844 (+0.36%)	0.285 (+0.06%)	0.270 (+0.06%)	0.269 (+0.07%)
C6	11.556 (+0.00%)	12.144 (+0.00%)	12.866 (+0.00%)	10.174 (+0.00%)	0.897 (+0.00%)	0.991 (+0.00%)	0.841 (+0.00%)	0.284 (+0.00%)	0.270 (+0.00%)	0.269 (+0.00%)

Table 13. Per-horizon mean MSE for MM-TSFLIB on Time-MMD. Aggregated over domains, seeds, and (for trained methods) backbones; $\Delta\%$ vs. C1 at the same horizon in parentheses.

Cond.	$H = 6$	$H = 8$	$H = 10$	$H = 12$	$H = 24$	$H = 36$	$H = 48$	$H = 96$	$H = 192$	$H = 336$
C1	19.746	20.173	20.547	15.708	0.925	1.041	0.920	0.486	0.502	0.471
C3	19.759 (+0.06%)	20.172 (-0.00%)	20.561 (+0.06%)	15.705 (-0.02%)	0.932 (+0.79%)	1.041 (+0.04%)	0.920 (+0.02%)	0.482 (-0.73%)	0.501 (-0.27%)	0.474 (+0.60%)
C4	19.766 (+0.10%)	20.140 (-0.16%)	20.564 (+0.08%)	15.705 (-0.02%)	0.928 (+0.34%)	1.038 (-0.26%)	0.914 (-0.61%)	0.484 (-0.26%)	0.500 (-0.45%)	0.474 (+0.64%)
C8	19.675 (-0.36%)	20.256 (+0.41%)	20.558 (+0.05%)	15.727 (+0.12%)	0.932 (+0.82%)	1.036 (-0.43%)	0.917 (-0.24%)	0.489 (+0.64%)	0.502 (+0.04%)	0.476 (+0.91%)
C9	20.136 (+1.97%)	20.690 (+2.56%)	21.047 (+2.43%)	16.093 (+2.45%)	1.032 (+11.58%)	1.146 (+10.11%)	0.991 (+7.73%)	0.479 (-1.33%)	0.490 (-2.42%)	0.469 (-0.56%)
C2	20.133 (+1.96%)	20.721 (+2.71%)	21.019 (+2.30%)	16.107 (+2.54%)	1.032 (+11.67%)	1.157 (+11.22%)	0.986 (+7.23%)	0.479 (-1.30%)	0.488 (-2.73%)	0.470 (-0.35%)
C5	20.127 (+1.93%)	20.733 (+2.77%)	21.020 (+2.30%)	16.111 (+2.57%)	1.034 (+11.81%)	1.156 (+11.06%)	0.988 (+7.48%)	0.481 (-0.95%)	0.488 (-2.71%)	0.469 (-0.44%)
C6	20.101 (+1.79%)	20.550 (+1.87%)	20.841 (+1.43%)	15.991 (+1.80%)	0.958 (+3.60%)	1.087 (+4.50%)	0.954 (+3.75%)	0.514 (+5.92%)	0.533 (+6.10%)	0.496 (+5.33%)

Table 14. Per-horizon mean MSE for TATS on Time-MMD. Aggregated over domains, seeds, and (for trained methods) backbones; $\Delta\%$ vs. C1 at the same horizon in parentheses.

Cond.	$H = 6$	$H = 8$	$H = 10$	$H = 12$	$H = 24$	$H = 36$	$H = 48$	$H = 96$	$H = 192$	$H = 336$
C1	19.256	19.494	19.883	15.081	0.843	0.906	0.747	0.298	0.302	0.285
C3	19.257 (+0.00%)	19.495 (+0.01%)	19.883 (+0.00%)	15.081 (+0.00%)	0.843 (+0.00%)	0.905 (-0.05%)	0.747 (+0.01%)	0.296 (-0.55%)	0.302 (+0.10%)	0.285 (-0.00%)
C4	19.257 (+0.00%)	19.495 (+0.00%)	19.883 (+0.00%)	15.082 (+0.00%)	0.843 (+0.00%)	0.905 (-0.08%)	0.746 (-0.02%)	0.296 (-0.54%)	0.302 (+0.18%)	0.285 (+0.02%)
C8	19.256 (-0.00%)	19.495 (+0.01%)	19.882 (-0.00%)	15.081 (+0.00%)	0.843 (+0.01%)	0.905 (-0.06%)	0.747 (+0.01%)	0.297 (-0.24%)	0.302 (+0.04%)	0.285 (+0.01%)
C9	25.118 (+30.44%)	25.516 (+30.89%)	25.670 (+29.11%)	19.717 (+30.74%)	1.678 (+99.13%)	1.776 (+96.13%)	1.393 (+86.63%)	0.430 (+44.30%)	0.384 (+27.04%)	0.389 (+36.24%)
C2	25.118 (+30.44%)	25.516 (+30.89%)	25.670 (+29.11%)	19.717 (+30.74%)	1.678 (+99.11%)	1.777 (+96.16%)	1.393 (+86.53%)	0.428 (+43.87%)	0.384 (+27.32%)	0.389 (+36.21%)
C5	25.116 (+30.43%)	25.516 (+30.89%)	25.670 (+29.11%)	19.717 (+30.74%)	1.678 (+99.11%)	1.777 (+96.15%)	1.393 (+86.59%)	0.428 (+43.73%)	0.384 (+27.30%)	0.388 (+36.16%)
C6	20.004 (+3.88%)	20.419 (+4.74%)	20.849 (+4.86%)	16.035 (+6.33%)	0.983 (+16.63%)	1.101 (+21.53%)	0.903 (+20.95%)	0.435 (+46.21%)	0.507 (+67.93%)	0.428 (+50.14%)